

LUNAR BUILD MISSION PLAN — a
Moon · 2026-06-03 · cut-fill balanced, optimized sequence

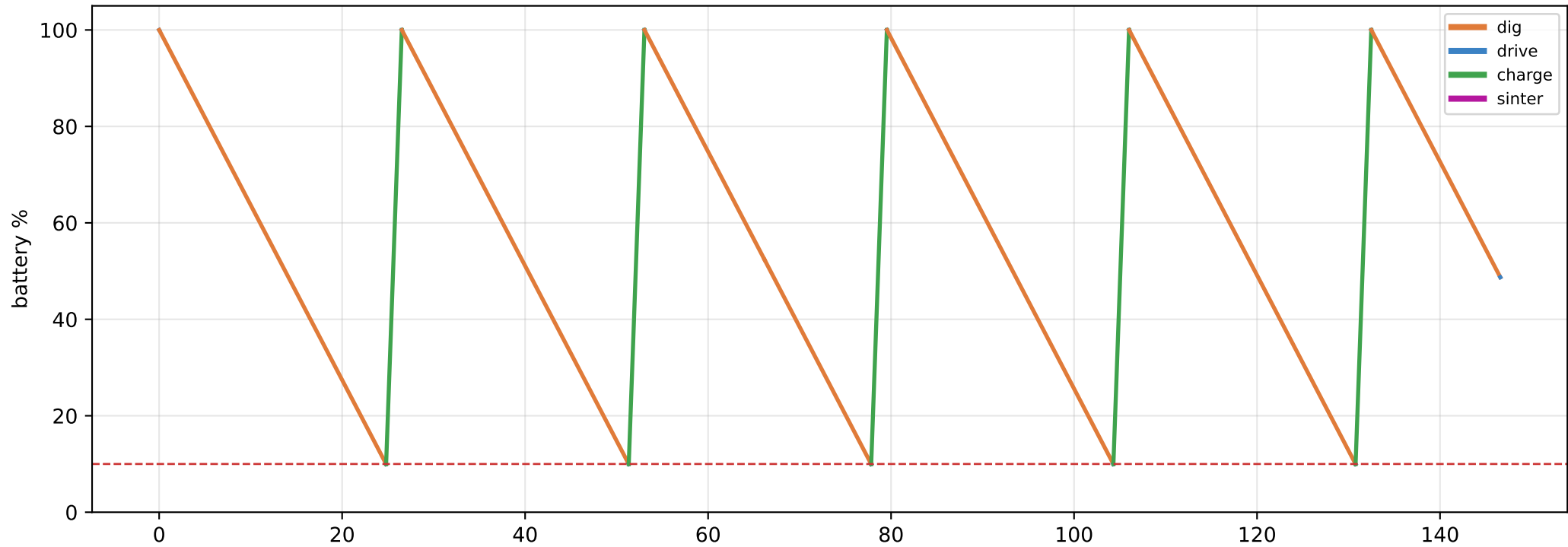
#	Trip	kind	Site (x,y)	Mass t	Duration	Energy (chg)
1	excavate → dump	cutfill	(5,5)	2.60	2.8 d	2.3
2	Excavate spoil: excavate	dig	(5,5)	3.16	3.4 d	2.7

MATERIAL BALANCE
cut 5.8 t → fill 2.6 t · surplus(spoil) 3.2 t · deficit(import) 0.0 t · sinter 0.00 t

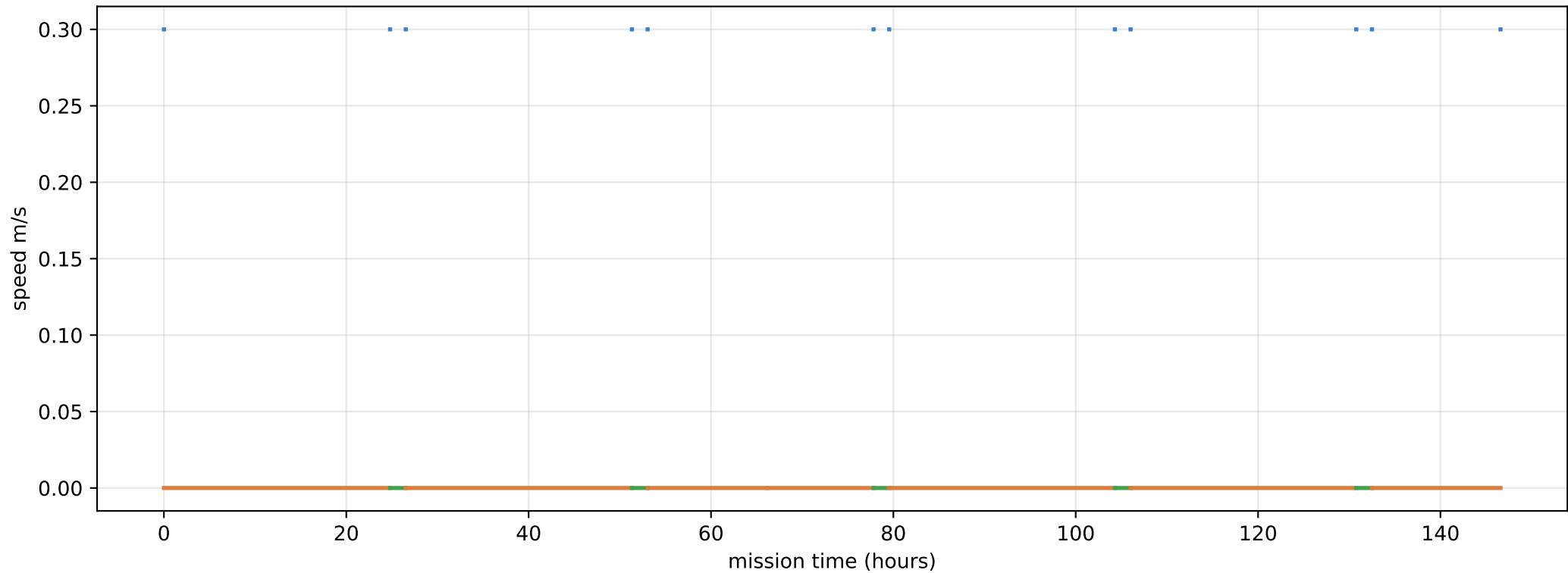
TOTALS project 6.1 d (147 h) moved 5.8 t (192 drum loads)
energy 24.0 MJ (5.0 charges, 5 recharge stops) drive 0.95 km
87 drum cycles; fill SENSED from motor current +/-2.6% (>half full), no load cell
per-sortie range 25 km (slope+slip @ 17 deg; 32 km flat)
1 lunar day ~ 30 Earth-days (354 h light)
ConOps: 70 km + 5-10 t / 11 d -> drive ~2.0 packs vs dig ~4-9 packs: drums dominate

Cut-fill balanced (excavated material routed to nearest fill; surplus→spoil, deficit→import). Grounded: per-body density/gravity (bodies.json); IPEx — 0.30 m/s, 42 kg/hr dig, 4151 J/kg, 135 J/m (slip-adjusted on a DEM), 4.79 MJ battery, 30 kg/drum; SINTER 0.92 MJ/kg [CALIB] (~220× dig). Recharge 700 W + sinter-head 1000 W are [CALIB]. Pluggable sequencer × objective; battery-aware mid-task recharge.

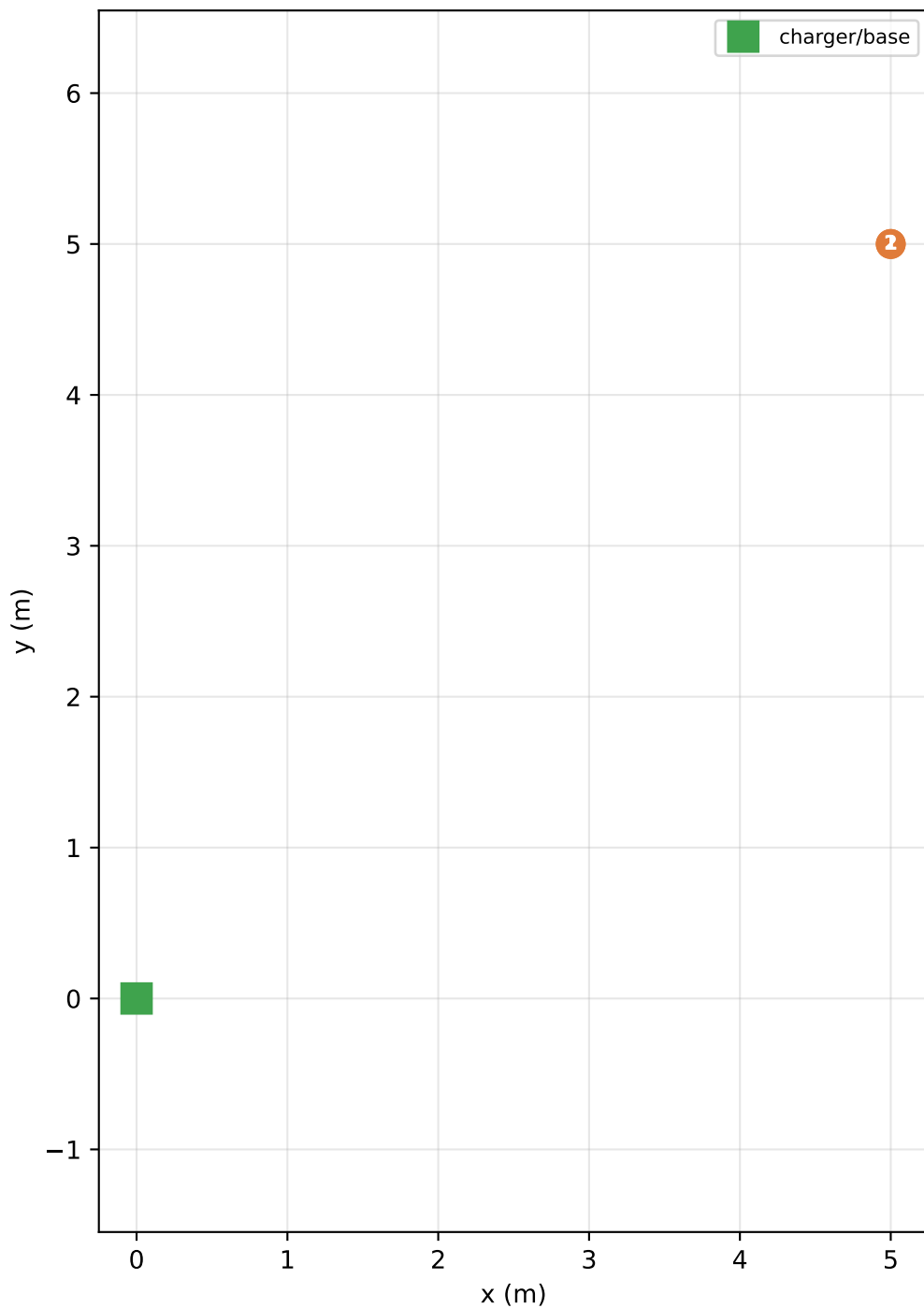
Battery draw over the planned project



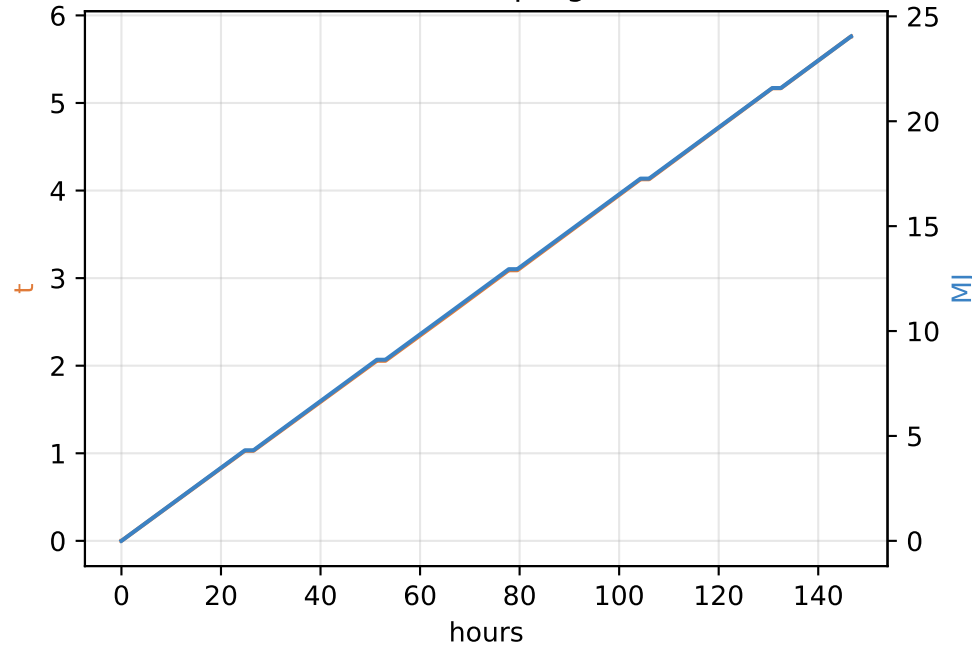
Speed profile



Site route + material flows (cut→fill)



Cumulative progress



Energy per trip (battery charges)

